

A Proposed New Methodology of Fault Prediction, Prevention and Optimization in a Multi-Source Smart Grid

Abstract- Smart electrical grids that incorporate multiple energy sources encounter threats from complexity, the intermittent operation of renewable energy sources, and uncertainties in grid operation. These factors lead to challenges in volume fault detection and prevention and dynamic optimization of grid operations. Traditional techniques suffer from slow fault detection times, erroneous detection (high false positives), and the inability to actively update or adapt the detection algorithms, removing adaptability from the system that directly impacts grid stability and efficiency. To overcome this limitation in this research the Leakage current monitoring applies a hybrid Particle Swarm Optimization-Genetic Algorithm (PSO-GA) model with an added adaptive filtering technique for early detection of insulation degradation. To discover fault states, an Isolation Forest-Support Vector Machine model optimized through Blue Whale Optimization (IF-SVM-BWO) is developed that forecasts transient and sustained faults with improved accuracy and fewer false positive incidents. Finally, fault avoidance utilized a novel approach to Generative Adversarial Networks, Gaussian Discriminant Analysis, and Fuzzy Logic (GAN-GDA-FLFD) approach to enable decision-making under uncertainty. Intelligent Protection Systems and proactive load control further mitigate cascading failures by isolating individual faults and managing demand conditions dynamically. The optimization of the power flow and voltage regulation is modelled through a Stackelberg game theoretical framework enhanced by a new Guide Waterwheel Plant Algorithm (Guide-WWPA) to rapidly adapt grid dynamics. If current regulation applications are using a Primal-Dual distributed averaging proportional-integral control scheme (PD-DAPI) to improve current performance rapidly and stably with less time delay.

Index Terms- Smart electrical grids, PSO-GA, IF-SVM-BWO, Guide Waterwheel Plant Algorithm, renewable energy sources, Guide-WWPA.

I. INTRODUCTION

The traditional electric grid (TEG), created over a century ago, faces significant challenges because of a huge increase in electricity use over the last few decades without the movement to upgrade its base infrastructure [1]. They have experienced unprecedented growth in usage, thus requiring broad-based, large-scale manipulation of consumption and production of electricity [2]. For example, worldwide energy demand increased by 4.6% in 2021 compared to 2020. The increase in demand leaves the electric grid (EG) with existing issues in providing power such as the previously mentioned power outages, load shedding, and inability to collaborate with other local grids [3,4]. These issues manifest within developing nations mainly because energy demand is increasing, while energy generation has fallen behind and the energy loss has compounded through increased loss due to inadequate management. For example, Pakistan should be considered a relevant case study that has an ongoing drastic increase in energy demand. Current electricity estimates indicate an overall

electricity demand of 115,000MW by 2030. However, the country has not generated enough energy and instead experiences large energy deficits (currently averaging about 6500MW, likely due to worsening in the future) [6].

For example, the system has seen its operational inefficiency measured as energy theft and average line losses of 18.7% (as a proportion of total generation) [7]. The electric grid also has severe operational issues regarding large demand with localized demand variations in hourly electrical loads that are primarily caused by the high variance placed on electrical loads such as air conditioning systems and electric-powered vehicles [8]. The rise of renewable energy resources like wind and solar is expected to make this situation more difficult and potentially result in further congestion or under-utilization of the electrical network [9]. Many of these renewables and loads are connected through power electronics converters. Many power electronics converters are now being used in grid-connected, distribution applications, not only for distributed power electronics converters but also "system level" power electronics converters within the electrical grid with various functions [10]. Although these types of converters have become an important part of the power grids, they can come with issues associated, including losing system inertia and causing sub-synchronous oscillations [11].

In a few recent studies, however, it has been shown that these issues can be reduced by modifying control strategies, and in turn, the controllability and flexibility of power converters can be fully utilized in the electrical grid [12]. A smarter energy grid opens a space to take advantage of energy storage providing flexibility and balance. It can reduce costs and improve efficiencies locally by creating better connections and management of distribution networks [13]. The faster integration of renewables into the market can result in quicker decarbonization of the grid, improved system operation security and efficiency of transmission and distribution as well as a decoupling of electric market prices providing a more reliable energy service [14]. Batteries disseminated efficiently in a smart power grid can maximize efficiency and provide fast flexible storage capacity [15].

A. Motivation & Objectives

Modern smart grids face significant challenges in ensuring reliable fault detection, prevention, and control due to system complexities and evolving operational demands. Important challenges in leakage monitoring, fault prediction, prevention, optimization, and control strategies must be overcome to improve stability and resilience in the grid.

- **Limitations in Leakage Current Monitoring:** Existing approaches rely on simulation data for the monitoring of leakage current, however, this may not

accurately reflect the actual conditions and affects the reliability of the fault detection.

- **Impact on Fault Prediction:** Previous research indicates that a lack of an integrated real-time monitoring platform creates obstacles to predictive fault identification, leading to delays in response and unanticipated failures. Moreover, existing fault prediction models reference predetermined attributes that are not necessarily full representations of real-time, near-term changes in the grid's activities.
- **Issues in Fault Prevention:** The efforts to prevent faults in existing approaches currently rely on mechanisms to assure that the grid will operate reliably under varying phenomena, leading to potential outages or blackouts. Without advanced data-driven methods for predictive fault mitigation, microgrids are at risk of experiencing unexpected disturbances. There is an increasing need for a stronger control framework to address the reliability and security of the grid under unpredictable fault-mitigating circumstances.
- **Problems in Optimization of Network Dynamics:** An existing reliance on predetermined optimization constraints reduces the responsiveness of power systems to near-term changes in demand and supply. Further, traditional optimization structures often do not adequately consider the impacts of renewable energy fluctuations on the overall dispatch efficiency of the system. More adaptive and dynamic optimization structures may lead to enhanced network resiliency.
- **Difficulties in Control Strategy:** Current approaches can create communication delays that can reduce control performance, potential instability, and slower response to large-scale power system disturbances. Managing the effects of communication delays is essential for preserving efficient transfer and reliable AGC operation in interconnected systems.

Motivated by the above main investigative issues, we suggested framework in this research applies to novel power systems and enhances performance, resilience, and adaptability to large-scale multi-sourced interconnected systems. The main objective of this research is given below:

- To develop a framework that collates and pre-processes data in real-time which allows for reliable and proactive failure detection.
- To develop a more advanced insulation monitoring process to detect early signs of failure and avoid system degradation.
- To leverage predictive maintenance to reduce system outages and improve fault detection.
- To streamline the distribution of power to enhance energy distribution dynamics and minimize losses during transmission.
- To improve control techniques to enhance grid stability, reduce latency, and improve them to enhance current-based control.

B. Research Contribution

The highlights of this research work are illustrated below;

- Developed a hybrid model for monitoring leakage current with neural networks using Particle Swarm Optimization and Genetic Algorithm (PSO-GA) for precise fault detection to support the prevention of insulation failure.
- Developed a framework for real-time fault prediction combining Isolation Forest, Support Vector Machine, and Blue Whale Optimization (IF-SVM-BWO) for improved anomaly detection and reduced false positives.
- Developed an intelligent fault prevention system with Generative Adversarial Networks, Gaussian Discriminant Analysis, and Fault Identification based on Fuzzy Logic (GAN-GDA-FLID) to actively prevent faults and improve microgrid resilience.
- Optimized power distribution dynamics through the Stackelberg game-theoretic framework and the Guide-Waterwheel Plant Algorithm (SGTF-Guide-WWPA) to improve energy flow regulation and reduce transmission losses.
- Enhanced control stability in large-scale grids by integrating Primal-dual and Distributed Averaging Proportional-Integral (PD-DAPI) protocols, to enhance current-based control.

C. Paper organization

This study's remaining sections are organized as follows: The survey of previous publications, including research gaps, is explained in Section II. The main issue with the current methods is described in Section III. The study methodology for the suggested model is presented in Section IV together with the pertinent mathematical representations, illustrations, and pseudocode. Provide the implementation details in section V. Section VI presents the experimental results and a comparison of the suggested and current methods. The proposed work's conclusion is addressed in Section VII.

II. LITERATURE SURVEY

The evolution of smart grids has encouraged the development of advanced techniques involving intelligent monitoring, data fusion, optimization algorithms, and deep learning for improved safety, fault detection, and energy forecasting.

In [16], an intelligent safety control framework for substations was proposed, integrating multi-sourced data and an Attention-Long-short Term Memory (LSTM) model for short-term spatial-temporal prediction. The system supports real-time monitoring via a 3D interface. However, high implementation costs and dependency on sensor reliability are key limitations. To enhance smart grid security, a fault detection mechanism based on edge computing and a safety risk map was presented

in [17]. The approach identifies critical nodes using potential functions, though its high computational demand can challenge resource-limited edge devices. A model predictive control (MPC)-based reactive power optimization method for voltage regulation in distributed energy resources was introduced in [18]. It utilizes event-triggered control for dynamic voltage band adjustment. However, real-time adaptation remains limited under fluctuating renewable energy scenarios.

Autonomous fault detection principles in smart grids were discussed in [19], outlining existing technical challenges and providing a case study. The research, however, lacks large-scale experimental validation and does not address long-term integration or scalability issues. A short-term load forecasting model combining an Improved Particle Swarm Optimization (IPSO) algorithm with LSTM was developed in [20]. It leverages multi-source meteorological and historical data but is constrained by computational complexity and data availability. In [21], a multi-source self-attention data fusion model combining LSTM and CNN was designed for False Data Injection Attack (FDIA) detection. While it enhances feature differentiation, its performance is affected by data synchronization and lacks evaluation against adversarial threats. The integration of Brain Emotional Learning with a Tilted Integral Derivative controller for Load Frequency Control was demonstrated in [22]. Although the model outperformed conventional methods, its feasibility under communication delays and cyber intrusions requires further validation. A Multi-Modal Multi-Task Transformer Network (M2TNet) was proposed in [23] for ultra-short-term wind power forecasting. It combines heterogeneous data sources and deep learning for improved accuracy. Nonetheless, training complexity and reliance on high-quality data are major limitations.

In [24], a deep learning-based electricity theft detection system was implemented using synthetic data and “principal component analysis (PCA)” and “minimum redundancy maximum relevance (mRMR)” for feature selection. While detection accuracy improves, the model's effectiveness depends on synthetic data reliability and is computationally intensive. A deep neural network-based approach, DeepOPF, was introduced in [25] for AC optimal power flow prediction using a prediction-reconstruction method. Although it offers computational advantages, it exhibits limited adaptability to unseen grid conditions. The energy forecasting solution in [26] uses ensemble and individual machine learning models trained on historical building data. The approach provides flexibility across multiple timescales but is constrained by time-consuming trial-and-error processes and data dependency. A

comparison of “Radial Basis Function Network (RBFnet), the Multi-Layer Perceptron (MLP), and a Convolutional Neural Network (CNN)” models for power flow prediction in unbalanced grids was carried out in [27]. Although accurate, the models are limited by the scope and variability of training data and their computational demand during deployment. Short-term load forecasting using deep learning was presented in [28], highlighting temperature and historical load as primary predictors. Despite high forecasting accuracy, limited historical data and the need for powerful computation hinder deployment. A voltage stability assessment framework employing a Transformer-based model and “Conditional Wasserstein GAN with Gradient Penalty (CWGAN-GP)” for rare event generation was proposed in [29]. Though capable of handling extreme cases, the model's high training complexity limits its scalability.

In [30], an electric vehicle (EV) charging power forecasting system was developed by integrating LSTM with reinforcement learning “adaptive exploration proximal policy optimization (AePPO)” and a Markov Decision Process. While it accounts for uncertainty effectively, it requires large data volumes and extensive computational

resources. A federated learning-based smart grid anomaly detection method was proposed in [31], enabling privacy-preserving local model training. However, its robustness in heterogeneous and adversarial environments has not been fully explored. An optimized PI controller for solar PV systems connected to the grid was introduced in [32], and validated on the IEEE 39-bus system under various conditions. Its practical applicability needs further exploration under more dynamic grid scenarios.

In [33], a defect detection method using “one-dimension convolution neural networks (1D-CNNs) and artificial neural networks (ANNs)” on sensor data was presented, achieving improved accuracy. Yet, it is susceptible to external noise and switching load interference. A multi-layer AI-based protection framework for medium-voltage distribution systems was developed in [34], combining communication-based and local protection algorithms. The system's complexity in adapting relay settings across varying grid configurations remains a challenge. Lastly, [35] proposed a hybrid model combining “Long Short-Term Memory and Adaptive Neuro-Fuzzy Inference System (ANFIS)” for smart grid fault detection using smart meter data. While the model enhances fault identification accuracy, its high time complexity restricts real-time implementation. Table 1 summarizes the work that has already been done.

III. PROBLEM STATEMENT

The numerous existing works and their associated responses are arranged in sequence of publication in this section. Furthermore, this study offers research solutions for the mentioned issues.

Authors in [36] research examines the prediction of faults in insulators through an evaluation of the insulator leakage current using different machine learning models such as “long short-term memory (LSTM), group method of data handling (GMDH), adaptive neuro-fuzzy inference system (ANFIS), bootstrap aggregation (bagging), sequential learning (boosting), random subspace, and stacked generalization

(stacking)” ensemble learning models are analyzed. The research method will also enhance these models with wavelet transform by achieving greater accuracy. The hybrid method the authors proposed substantially bolsters previous machine learning forms making them useful for various situations. This research looks at several important technologies related to equipment operation and maintenance in power systems (PS) with a focus on intelligent maintenance and fault diagnosis [37]. The case study for the research is the Longyangxia Hydropower Station to examine the complementary features of solar and hydro energy and transformers. The study also looks into different advanced materials (e.g. amorphous alloys) to enhance transformer performance. The paper concludes with a long-action maintenance strategy to increase the reliability of the components. Some of the problems detected in these papers are:

- There are limitations to the study. The first is the reliance on simulation data, which may not sufficiently capture real-world conditions. Additionally, models may require additional validation studies in different settings. Lastly, focusing on only leakage current may cause other influencing factors on insulator performance to be neglected.
- The problem identified is the Lack of a common platform for real-time monitoring of electrical equipment, maintenance, and predictive fault detection is a significant challenge in modern power systems. Failures can occur without warning if operators do not have access to sophisticated data mining and analytics. Failure detection can go unnoticed or take too long to detect, resulting in sudden unplanned failures.

In this research [38], they present a Sensor Fault Tolerance methodology for hierarchical control in independent microgrids that utilize a modified Kalman filter to increase reliability and safety. This methodology provides a reliable means of reducing hardware redundancy while yielding the same control outputs under sensor fault conditions while maintaining voltage within acceptable limits. They validated the methodology using simulation in a MATLAB/ Simulink product framework that allowed us to evidence the stability of the system in maintaining microgrid operation. The study [39] presents, “multi-layer perceptron-Extreme Learning Machine (MLP-ELM)” a machine-learning decision support system for estimating total stability in smart-grid networks. The MLP-ELM generates stability forecasts, and in the development phase, PCA was utilized in the MLP-ELM effective feature extraction process. The study results also include comprehensive empirical evaluations against existing methods to support the vision. The outcome of the MLP-ELM implementation highlights the effectiveness of the proposed smart-grid network stabilization solution. This study presents a decentralized Model Predictive Control (DMPC) approach to advance automatic generation control (AGC) in wind-storage systems [40]. By utilizing an adaptive fuzzy-based coordinated control strategy, AGC demand is allocated to the wind storage technologies and the reliability of the wind farms is improved by satisfying the

operational scenarios. A simulation for a system with three areas was conducted to validate the performance enhancement in AGC. The approach discussed in this researchers’ study introduces an effective interface system between stored wind power and thermal plants for maintaining secure grid operation. Several issues identified in this research include:

- Within these systems, there are a finite number of strong aides to ensure stability against these blackout conditions. A robust control strategy will be valuable in enhancing MG's reliability and security.
- One limitation of the research is the predefined features which may fail to capture the complete real-time oscillations within smart grid networks. The MLP-ELM has not been validated in highly dynamic or extreme conditions.
- With the growing pace of grid interconnections, AGC will use open communication networks to carry out control signals to wide areas thereby introducing time delays to AGC performance. These time delays in AGC performance will reduce control performance, and introduce instability and time delays in response time. It is critical to address the overall impact of communication time delays, to improve AGC performance in a large-scale power system setting.

A superior hybrid algorithm combining particle swarm optimization and genetic algorithm (PSO-GA)- is proposed, enhancing the accuracy of leakage current prediction for insulators. The utilization of the Isolation Forest combined with a support vector machine with blue whale optimization (IF-SVM-BWO) algorithm for real-time fault detection in the power system. Generative Adversarial Networks, Gaussian Discriminant Analysis, and Fuzzy Logic-Based Fault Detection (GAN-GDA-FLFD) are strong solutions to enhance microgrid stability while exhibiting resiliency against disturbance. To improve real-time behavior, the Guide-Waterwheel Plant Algorithm (Guide-WWPA) has been used and can adjust power flow and voltage regulations in real-time based on variations in the smart grid. Finally integrating the novel Stackelberg game-theoretic framework providing outputs with the Guide-Waterwheel Plant Algorithm into a viable and reasonable distribution with the proposed "SDTF-Guide-WWPA" for optimizing overall stability in smart grids. Finally, the Primal-dual and distributed averaging proportional-integral (PD-DAPI) protocols are proposed to control the counteract time delay, current optimization with reduced time delay, and improved stability.

IV. PROPOSED METHOD

The methodology proposed intends to address fault detection, prevention, and optimization in a multi-sourced smart electrical grid by using advanced monitoring, prediction analytics, and control methods. By including real-time data collection, sophisticated predictive monitoring of faults, proactive prevention of faults, and optimization of the electrical grid network in operation. This approach will assist in the

improved reliability, stability, and resiliency of the smart grid against faults and operational uncertainties. The overall architecture of this research is shown in Fig. 1.

- Real-time data collection and preprocessing

- Leakage Current Monitoring
- Fault Prediction
- Fault Prevention
- Optimization of Network Dynamics
- Control Strategy

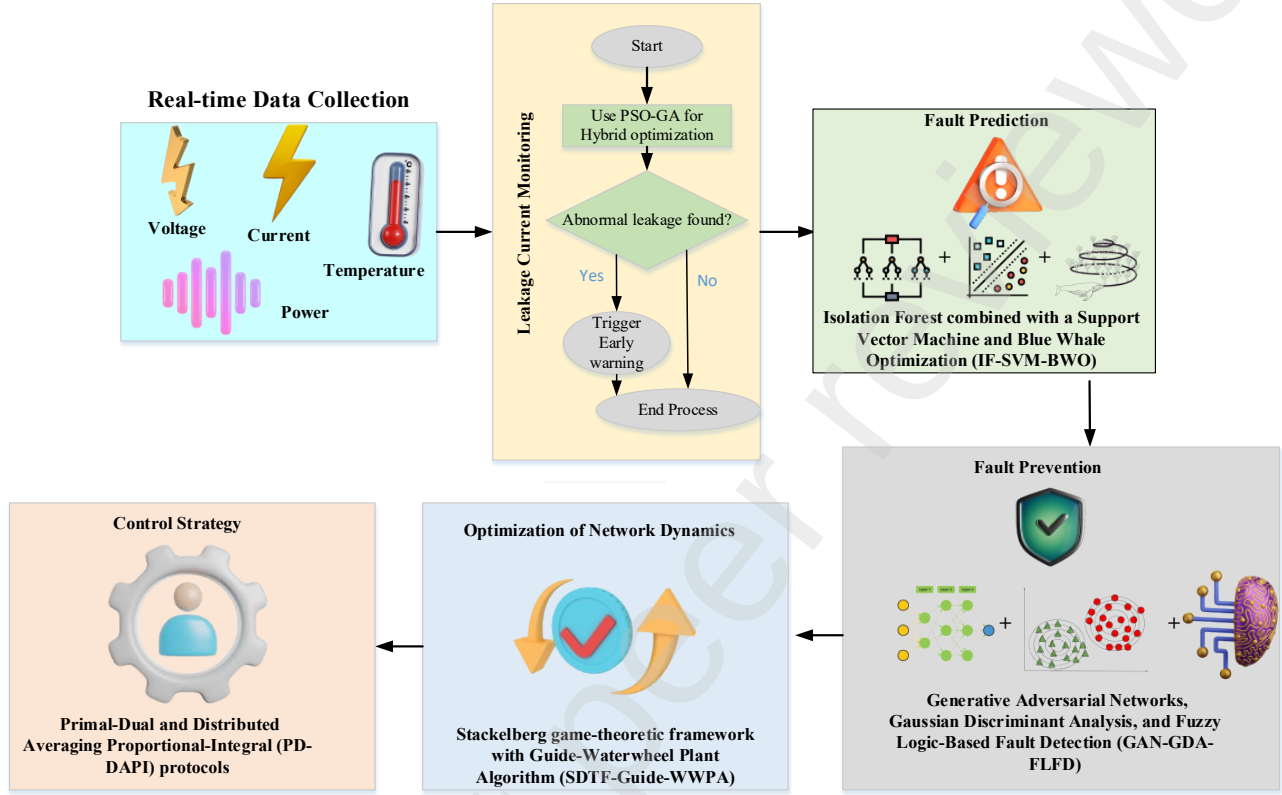


Fig. 1 Overall architecture of this research

1. Real-time data collection and preprocessing

This stage includes the collection of time series data on overloaded quantities, including voltage, current, frequency, and environmental conditions about energy sources. Historical operation data is also considered to help understand typical and atypical behaviors of the system. Data preprocessing methods are then applied to the data to improve data quality (e.g., noise reduction and normalization). Outlier detection methods statistical, machine learning, etc. are then employed to define outliers and work to correct missing data values. Feature selection methods established relevant inputs to analysis. The approach is aimed at delivering a high-quality dataset for enhanced fault prediction/prevention accuracy.

Real-time monitoring on the electrical grid will be developed being that it is possible to continuously capture time-series measurements of electrical parameters, such as voltage $V(t)$, current $I(t)$, frequency $f(t)$ and relevant environmental parameters, such as temperature $T(t)$, humidity $H(t)$ and irradiance $R(t)$ from renewables.

We will also include historical operational data, such as fault records and maintenance records that provide contextual

information to discriminate between typical system behavior from atypical behavior. A data preprocessing pipeline is applied to verify the quality of the data and its appropriateness for further analysis. Initially, we filter the raw sensor measurements for noise. The aim is to capture the phenomena of interest by removing random high-frequency noise induced by several external interactions, such as environmental interference and measurement inaccuracies, by using mathematical filters. We can use moving averages and filters to smooth or flatten the signal by attenuating high-frequency noise, where mathematically it can be represented as:

$$\hat{x}_t = \frac{1}{2\ell + 1} \sum_{i=-\ell}^{\ell} x_{t+i} \quad (1)$$

Where $\hat{x}_t$ is estimated/smoothed data, x_t is the raw measure at time $[t]$, and ℓ is the window size that controls smoothing.

After reducing noise, normalization is then applied to standardize scales of the features to account for greater magnitude differences across variables. Specifically, Z-score

normalization transforms each feature using as given by the following expression:

$$\mathbb{x}_t^{norm} = \frac{\mathbb{x}_t - k}{\varphi}, \quad (2)$$

Where k and φ are the mean and standard deviation computed from the training dataset. Missing or corrupted data is dealt with in a two-pronged approach to imputation. Simple imputation techniques replace lacking values with the mean or median. Complex implementation techniques such as k-nearest neighbor (kNN) estimate missing values by averaging the neighboring samples as given by the following expression:

$$\mathbb{x}_t^{imp} = \frac{1}{\sigma} \sum_{i=1}^{\sigma} \mathbb{x}_{t_i} \quad (3)$$

Where $\{\mathbb{x}_{t_i}\}$ is the set of the σ valid observations closest to the missing value at t .

Outlier detection is applied for finding measurements that are aberrant which may imply either a sensor failure or a transient disturbance. The Z-score is a common statistic that quantifies deviations from normal behavior as given by the following expression:

$$Z_t = \frac{\mathbb{x}_t - k}{\varphi}, \quad (4)$$

A threshold τ is set such that observations with $|Z_t| > \tau$ are flagged as outliers for removal or correction. To further develop the predictive performance, the pre-processed data undergoes feature extraction to extract informative features that contain underlying fault signatures. Features that employ time-domain statistical measures, variance, skewness, and kurtosis are computed over sliding windows and capture temporal variations as given by the following expression:

$$k = \frac{1}{N} \sum_{i=1}^N \mathbb{x}_i \quad (5)$$

Where N is the length of the window. Frequency-domain features are measured using the Fast Fourier Transform (FFT); this process quantifies the spectral energy in a defined frequency E_f bands:

$$E_f = \sum_{f=f_1}^{f_2} |\mathfrak{X}(f)|^2 \quad (6)$$

If $\mathfrak{X}(f)$ is the Fourier Transform coefficient between frequency f_1 and frequency f_2 . Wavelet transforms can also be used to capture transient and localized fault patterns, with a multi-resolution analysis.

To simulate thermal dynamics in the grid, temperature $T(t)$ is modelled as a linear function of voltage and current measurements as given by the following expression:

$$T(t) = 20 + 0.5 * \mathbb{V}(t) + 0.3 * \mathfrak{I}(t) \quad (7)$$

Where the coefficients approximate the resistive heating and dielectric losses found in power components (e.g. transformers and transmission lines) under normal operating conditions. Thus, this empirical relationship makes thermal monitoring simple while providing a reasonable representation of the general trends that can be used for fault detection. For greater fidelity, nonlinear or data-driven models could be used instead in actual deployments.

Afterwards, a dimensionality reduction technique such as Principal Component Analysis (PCA) transforms the feature space into principal components:

$$\mathbf{z} = \mathbf{W} \quad (8)$$

The cleaned feature set is used to train supervised learning models aimed at all fault classification and prediction. Continuous variables (voltage, current, temperature) were discretised into fault/no-fault classes using:

Threshold: $\mathbb{V}(t) < 200V$ or $\mathfrak{I}(t) > 5A \rightarrow \text{fault (class 1)}$.

Isolation forest: detected outliers (5% of samples) as faults.

2. Leakage Current Monitoring

Monitoring leakage current is an important step for identifying insulation degradation in power systems. A hybrid model implemented utilizing Particle Swarm Optimization and Genetic Algorithm (PSO-GA), Real-time sensors will continuously monitor variations in leakage current in insulators and power components. When variations become abnormal, the system will provide early warning and prevent system failure.

Adaptive filtering methods are proposed to remove noise from sensor readings. Using trained-based predictions will be useful to isolate regular fluctuations, cycle-to-cycle variations, and at-risk leakage fluctuations. This proactive measure will be helpful in the longevity and reliability of the electrical infrastructure. We can model the leakage current $\mathfrak{I}_L$, as a submission signal composed of harmonic elements and additive noise, given in the time-domain as:

$$\mathfrak{I}_L = \sum_{n=1}^N A_n \sin(2\pi f_n t + \phi_n) + \eta(t), \quad \eta(t) \sim N(0, \mu^2) \quad (9)$$

where A_n , f_n and ϕ_n are the amplitude, frequency, and phase of the n^{th} harmonic respectively, and $\eta(t)$ denotes Gaussian noise with zero mean and a variance of μ^2 . The harmonic components stem from the periodic distortion of grid voltage loads and the grid operational conditions while the noise is due to electromagnetic interference (EMI) and sensor noise.

To extract the true leakage current we use the following expression:

$$S(\mathfrak{I}_L) = \sum_{n=1}^N A_n \sin(2\pi f_n \mathfrak{I} + \phi_n) \quad (10)$$

An adaptive Least Mean Squares (LMS) filter is applied. The LMS filter performs by adjusting its weights to reduce the error between the input $\mathfrak{I}_L$, which contains noise, and a reference signal $d(\mathfrak{I})$, based on the previous noise history. The filtered output $y(\mathfrak{I})$ is:

$$y(\mathfrak{I}) = W^t(\mathfrak{I}) x(\mathfrak{I}) \quad (11)$$

Where W^t is the time-varying weight vector, and

$x(\mathfrak{I})$ represents the input signal vector with filter length l .

The error signal $e(\mathfrak{I})$ is calculated as;

$$e(\mathfrak{I}) = d(\mathfrak{I}) - y(\mathfrak{I}) \quad (12)$$

The weights are updated iteratively using the LMS rule:

$$W(\mathfrak{I} + 1) = W(\mathfrak{I}) + \sigma e(\mathfrak{I}) x(\mathfrak{I}) \quad (13)$$

Where σ represents the step size parameter which balances the speed and stability of converging. Smallest values of σ will ensure stability but slow convergence, larger values will cause divergence.

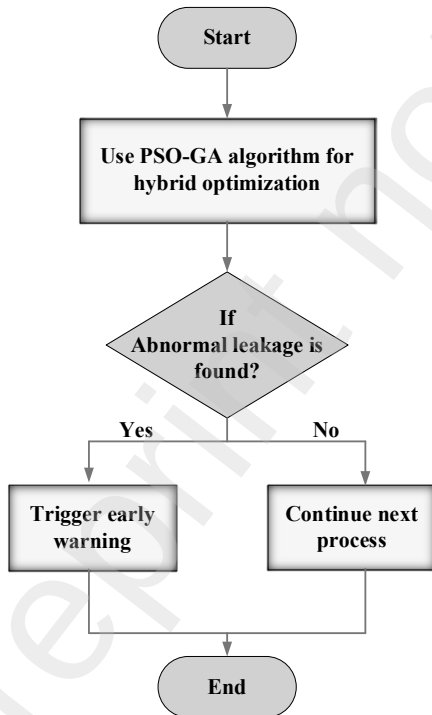


Fig. 2 Flow diagram for Particle Swarm Optimization and Genetic Algorithm (PSO-GA)

The hybrid Particle Swarm Optimization-Genetic Algorithm (PSO-GA) is proposed to optimize the parameters critical to our method, LMS filter weights and dynamic detection thresholds. The flow diagram for Particle Swarm Optimization and Genetic Algorithm (PSO-GA) is shown in Fig. 2.

In PSO, each particle q designates a future solution using the position p_q and velocity α_q . The velocity and position updates for PSO are defined as:

$$\alpha_q(\mathfrak{I} + 1) = w\alpha_q(\mathfrak{I}) + c_1 r_1(p_q^{\text{best}} - p_q(\mathfrak{I})) + c_2 r_2(g^{\text{best}} - p_q(\mathfrak{I})) \quad (14)$$

$$p_q(\mathfrak{I} + 1) = p_q(\mathfrak{I}) + \alpha_q(\mathfrak{I} + 1) \quad (15)$$

In this case, w is the inertia weight, c_1 and c_2 are cognitive and social acceleration coefficients, and $r_1, r_2 \sim \mathbb{U}(0, 1)$ are random values. The personal best and global g^{best} are there to direct particles toward the best solutions. GA enhances population diversity through selection, crossover, and mutation. The selection probability for the $\mathbb{P}_i$ individual is:

$$\mathbb{P}_i = \frac{f_i}{\sum_{j=1}^N f_j} \quad (16)$$

Where f_i and f_j are the fitness of the i, j^{th} individual. Crossover blends two parents to generate offspring (Q):

$$Q = \beta \text{Parent}_1 + (1 - \beta) \text{Parent}_2, \quad (17)$$

where $\beta \in [0, 1]$ is a blending coefficient. Mutation introduces diversity via Gaussian perturbation:

$$p'_q = p_q + N(0, \delta_m) \quad (18)$$

With δ_m controlling mutation magnitude.

3. Fault Prediction proposed

For real-time fault detection, an Isolation Forest (IF) with a Support Vector Machine (SVM) was proposed. To optimize the SVM parameters, the Blue Whale (BWO) Optimization is utilized to improve the accuracy of the fault prediction (Fig. 3).

Using the sensors and application, the model will monitor grid dynamics status on 3 main parameters: the power flow, frequency and voltage stability.

The IF-SVM-BWO model allows for the classification of transient faults from sustained faults for the improvement in the algorithm rendering a reduced false positive. As the model is real-time fault detection, the notifications are sent to the grid operator, for them to take further action for the system to react to a full-blown fault. This enables an improved, condition-based maintenance practice that has the potential for reduced

unexpected downtime that occurs due to system interruptions. These processes assist in improving the stability of the grid and better fault response. Furthermore, the history of the faults,

which can be worked into real-time monitoring, can provide reliability and flexibility of the services to adapt between the historical and present states of the grid.

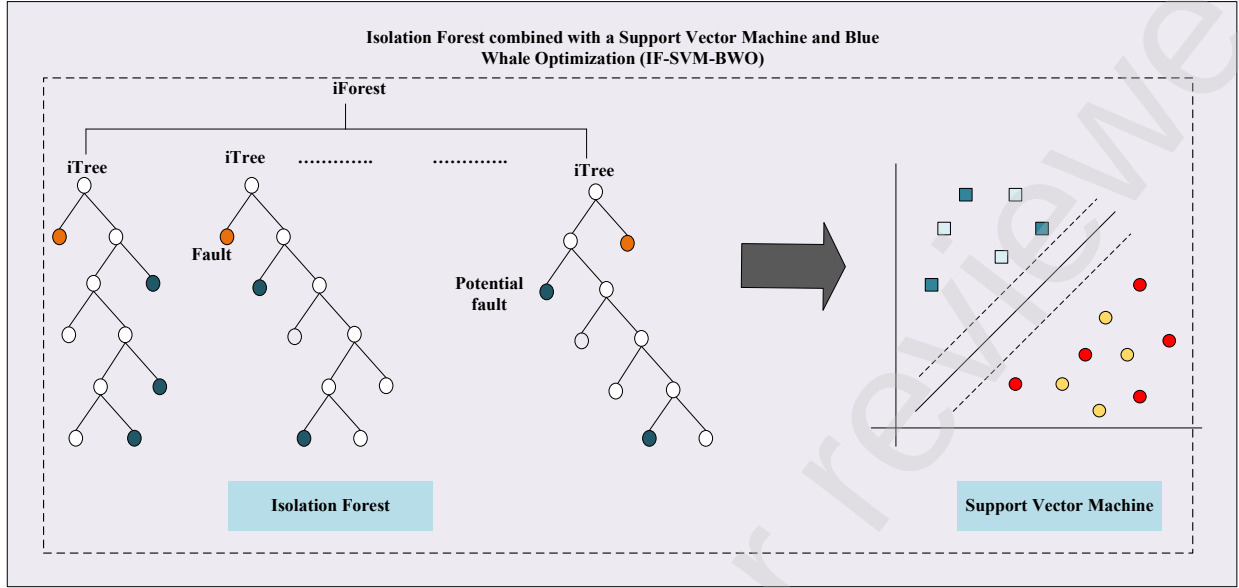


Fig. 3 Fault Prediction using IF-SVM-BWO

Isolation Forest is based on the idea of isolating anomalies without profiling normal data points. In a dataset $\mathcal{D} = \{x_1, x_2, \dots, x_n\}$, where each $x_i \in \mathbb{R}^d$ are built by recursively partitioning the data space into isolation trees. The measure of isolation quality is given by the average path length $h(x)$, in which it takes to isolate a sample x . The expected average path length for a sample size ζ is approximated as,

$$\zeta = 2\zeta(\zeta - 1) - 2\frac{\zeta - 1}{\zeta} \quad (19)$$

Where the harmonic number $\zeta(i)$ is approximated by,

$$\zeta(i) = \ln(i) + \gamma \quad (20)$$

With $\gamma \approx 0.5772$ being Euler constant. This provides the foundation for computing an anomaly score which is given by the following expression.

$$Si(x, \zeta) = 2^{-\frac{h(x)}{\zeta}} \quad (21)$$

where $Si(x, \zeta)$ near 1 suggests a strong anomaly, therefore successfully detecting early signs of faults concerning distinct parameters in the electrical grid, namely active power $Power(t)$, frequency $f(t)$, and voltage magnitude $V(t)$. After anomaly detection, the found instances are classified using a Support Vector Machine. SVM makes a hyperplane that best separates classes in this case transient and sustained faults in a transformed high-dimensional feature space. Given a training

set $\{(X_i, Y_i)\}$ with labels $y_i \in \{-1, +1\}$, SVM minimizes the regularized risk functional

$$\min_{W, \mathbb{b}, \xi} \frac{1}{2} \|W\|^2 + MC \sum_{i=1}^n \xi_i \quad (22)$$

$$y_i(W \phi(x_i) + \mathbb{b}) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad (23)$$

W is the weight vector, $\mathbb{b}$ is the bias term, ξ slack variables allow for soft margins, MC is a penalty parameter balancing maximization of margin with tolerance for misclassification. The kernel trick uses the Radial Basis Function kernel to map inputs into a higher-dimensional space.

$$Kl(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (24)$$

Where γ controls the spread of the kernel. The dual optimization problem

$$\max_{\psi} \sum_{i=1}^n \psi_i - \frac{1}{2} \sum_{i,j=1}^n \psi_i \psi_j y_i y_j Kl(x_i, x_j) \quad (25)$$

$$0 \leq \psi_i \leq MC, \quad \sum_{i=1}^n \psi_i y_i = 0 \quad (26)$$

Yields Lagrange multipliers ψ that define the classifier's decision function.

$$f(x) = \text{sign}(\sum_{i=1}^n \psi_i y_i Kl(x_i, x_j) + \mathbb{b}) \quad (27)$$

The sensitivity of SVM performance to the hyperparameters HY_p and blue whale optimization is incorporated to efficiently

explore the parameter space. BWO emulates the social hunting behavior of humpback whales via encircling prey, bubble-net feeding, and searching mechanisms. The encircling step updates candidate solutions according to,

$$\mathbb{D} = |HY_p \cdot X^*(t) - X(t)|, \quad X(t+1) = X^*(t) - A \cdot \mathbb{D}, \quad (28)$$

Where $X^*(t)$ denotes the current best solution, $X(t)$ a candidate, and the coefficient vectors

$$A = 2LD \cdot RV - LD, \quad HY_p = 2 \cdot RV \quad (29)$$

With LD linearly decreasing from 2 to 0 and RV a random vector in $[0,1]$. The bubble-net feeding spiralling behavior is modelled by,

$$X(t+1) = \mathbb{D}' e^{bl} \cos(2\pi l) + X^*(t) \quad (30)$$

Where,

$$\mathbb{D}' = |X^*(t) - X(t)| \quad (31)$$

b controls the logarithmic spiral shape, and $l \in [-1, 1]$ is random. Exploration is promoted by

$$X(t+1) = X_{rand} - A \cdot |HY_p \cdot X_{rand} - X(t)| \quad (32)$$

Where X_{rand} is a randomly chosen candidate from the population. This evolutionary process minimizes the objective.

$$F(HY_p, \gamma) = 1 - Accuracy(HY_p, \gamma), \quad (33)$$

The accuracy is evaluated on a validation dataset. Real-time monitoring is achieved by continuously acquiring the feature vector.

$$x(t) = [Power(t), f(t), V(t)], \quad (34)$$

Feeding into the isolation forest to compute $Si(x(t), \zeta)$ anomaly scores exceeding a threshold trigger classification by the optimized SVM, which determines fault types. When sustained faults are detected, the system issues immediate alerts, enabling proactive maintenance. The pseudocode for the proposed IF-SVM-BWO is illustrated in algorithm 1.

Algorithm 1: Fault Prediction using IF-SVM-BWO

Input: Real-time grid dynamics data (power flow, frequency stability, voltage stability) $\mathcal{D} = \{x_1, x_2, \dots, x_n\}$, where $x_i \in \mathbb{R}^d$

Output: Classification of faults (transient or sustained) and notification to grid operator.

1. Build Isolation Trees:
2. **For** $i = 1$ to num_{trees} :
3. Create a random subset of $\mathcal{D}$

4. Recursively partition the subset until each data point is isolated
 5. Calculate the average path length $h(x)$ for each data point x in $\mathcal{D}$
 6. Anomaly Score:
 7. Compute the anomaly score for each x in $\mathcal{D}$ using $h(x)$
 8. Train SVM:
 9. Use training data
 10. Optimize SVM parameters using Blue Whale Optimization (BWO)
 11. BWO Parameter Optimization:
 12. Initialize a population of blue whale positions
 13. While ($iteration < max_{iterations}$):
 14. Calculate the fitness of each whale
 15. Update whale positions based on BWO rules
 16. Update best whale position
 17. End While
 18. Set SVM parameters to the best whale position
 19. For each new data point x_{new} :
 20. Calculate anomaly score using Isolation Forest
 21. Classify x_{new} using the trained SVM
 22. If classified as fault:
 23. Determine fault type (transient or sustained)
 24. Send notification to grid operator
 25. End
-

Temperature $T(t)$ is an important fault classification label because it can be sensitive to abnormal grid action. Generally, overheating may mean that:

Insulation is being destroyed: Local temperature is rising because we have increased leakage current.

Overloads: Resistive Losses (I^2R) increase nonlinearly with current, therefore causing heating.

Cooling Failures: This is the case for transformer and inverter failures/ faults. Thus, when we include temperature into anomaly score $Si(x, \zeta)$, we leverage the fact that the model gives higher scores to thermal anomalous events that are leading indicators of incipient faults.

4. Fault Prevention proposed technique

To reinforce the resilience of microgrids, Generative Adversarial Networks, Gaussian Discriminant Analysis, and Fuzzy Logic Fault Detection (GAN-GDA-FLFD) are proposed. The GAN methodology generates synthetic faulty situations to provide a stronger foundation to the control models, while the GDA application classification states the system in the application of faults, including key fault indicators, voltage inconsistencies, and load imbalances.

The newly integrated FLFD methods improve decision-making by considering uncertainty in fault identification to improve robustness in adaptive conditions. The control models, along

with the new methodologies, provide additional proactive fault mitigation approaches while modeling various operational conditions while also exposing latent vulnerabilities within the power networks.

Additionally, Smart Protection Systems are applied to quickly identify the fault condition and respond accordingly with systems designed to isolate faulty equipment, automatic load shedding, and real-time fault containment models to reduce the risk of an immediate cascading failure. Proactive Load Management methods are also used to manage demand-side strategies to dynamically prescribe energy consumption considering the state of the network, such as load shedding to ensure that robust distribution minimizes jeopardized power grid conditions.

a. Synthetic fault scenario generation using GAN

The initial stage in fault prevention is to create varied fault conditions which is imperative for training a good detection model. This is done by using Generative Adversarial Networks (GANs) which comprise a generator G and a discriminator D . The role of the generator is to create realistic fault data from random noise, whilst the discriminator is to try to tell what is real and what is generated. The adversarial training objective is:

$$\min_G \max_D \text{Loss}_{GAN} = E_{x \sim p_{data}}[\log D(x)] + E_{z \sim p_z}[\log(1 - D(G(z)))] \quad (35)$$

To enhance stability during training, we adopt the least-squares loss function for the discriminator and generator respectively:

$$\text{Loss}_D = \frac{1}{2} E_{x \sim p_{data}}[(D(x) - 1)^2] + \frac{1}{2} E_{z \sim p_z}[(D(G(z)))^2] \quad (36)$$

$$\text{Loss}_G = \frac{1}{2} E_{z \sim p_z}[(D(G(z)) - 1)^2] \quad (37)$$

These ensure that the synthetic samples generated by G closely approximate the distribution of real fault conditions.

b. Gaussian Discriminant Analysis

A Gaussian distribution is fitted to every class of data via the probabilistic data categorization technique known as GDA. Let n be the number of classes, and let $b \in \{1, 2, \dots, n\}$ stand for the classes. Given class $b = k$, the potential of sampling an observation a is,

$$F(a|b = k) = N(a; \alpha_k, \rho_k) \quad (38)$$

where the Gaussian distribution fitted to class k has a mean vector α_k and a covariance matrix ρ_k . The class of the

observation is inferred by the GDA classifier using the Bayes theorem:

$$F(b = k|a) = \frac{F(a|b = k)F(b = k)}{\sum_{i=1}^n F(a|b = i)F(b = i)} \quad (39)$$

To optimize the distance between classes, GDA can be utilized as well for supervised dimensionality reduction, by extending the data to a smaller subspace. Assuming that every class has the same covariance matrix $\Sigma_b = \Sigma$ for $b = \{1, 2, \dots, n\}$, we can select this reduced subspace to be linear, which facilitates the interpretation of GDA findings. The resultant algorithm is known as Linear Discriminant Analysis (LDA). The following formula is obtained by taking the log-posterior of eq (38).

$$\log F(b = k|a) = (\Sigma^{-1} \alpha_k) a + U \quad (40)$$

The significance of the characteristics of the data in distinguishing the classes is determined by the coefficients in $\Sigma^{-1} \alpha_k$ (U contains additional irrelevant factors). Unimportant characteristics can be disregarded below a specific point to lower the problem's dimensionality.

The election of Σ often arises under ambiguous or nonlinear operating conditions. To model this Fuzzy Logic-Based Fault Detection (FLFD) is proposed. Each variable is mapped to fuzzy sets using sigmoid membership functions:

$$\mu_A(x) = \frac{1}{1 + e^{-\alpha(x-c)}} \quad (41)$$

Fuzzy rules then infer the degree of fault risk:

$$\mu_{out}(y) = \max_i \min_j |\mu_{A_i}(x_1), \mu_{B_j}(x_2), \dots| \quad (42)$$

The fuzzy output is converted to a crisp decision using the centroid method:

$$y^* = \frac{\int y \cdot \mu_{out}(y) dy}{\int \mu_{out}(y) dy} \quad (43)$$

This approach improves the interpretability and robustness of incomplete or noisy inputs. To integrate the fuzzy and the probabilistic and fuzzy interactors, a Fault Risk Index (FRI) is a weighted combination of system-level indicators:

$$FRI = \alpha_1 \cdot \varphi_V + \alpha_2 \cdot \Delta Power + \alpha_3 \cdot \lambda + \alpha_4 \cdot P_f \quad (44)$$

Where:

- φ_V : standard deviation of voltage magnitude.
- $\Delta Power$: deviation between measure and forecasted power.

- P_f : predicted fault probability from GDA.
- $\alpha_1, \alpha_2, \alpha_3, \alpha_4$: weighting factors

When the FRI exceeds a set threshold, preventive actions are initiated:

$$FRI(t) \geq \theta_{ALAM} \quad (45)$$

$$Load_t^{new} = Load_t \cdot (1 - \beta \cdot FRI) \quad (46)$$

This fault prevention framework introduces a rigorous multi-layered model that merges probabilistic classification, fuzzy reasoning, real-time control, and optimization. By evaluating fault likelihoods, qualifying operational deviations and dynamically adjusting system responses, it enables high-fidelity fault prevention with improved stability and resilience in multi-score smart grids.

5. Optimization of Network Dynamics

The interactions between power generators (leaders) and microgrids (followers) are well captured by the Stackelberg game-theoretic framework. In this framework, the generators develop power generation levels that maximize the power system's overall efficiency. The microgrids react to the generated power by adapting their consumption and storage to achieve stability. However, the adaptability to changes in demand and supply on a real-time basis is limited to the constraints imposed a priori.

A Novel Guide-Waterwheel Plant Algorithm (Guide-WWPA) is proposed to overcome these challenges. The algorithm changes the power flow and voltage regulation adaptively to account for changes in the grid and can provide solutions to the issues discussed previously. The power distribution decision-making enhances the welfare of the system by lowering the power transmission loss. The algorithm will reliably maintain a stable and efficient power supply concerning changes in the grid. In conclusion, we have put forward a novel **Stackelberg game-theoretic framework with the Guide-Waterwheel Plant Algorithm (SDTF-Guide-WWPA)** to improve the reliability of the decision tool to maintain the stability of the network.

Consider a power grid composed of $N + 1$ buses, where the bus θ_0 is the Slack reference, and the remaining N buses are indexed as $\mathcal{B} = \{\theta_1, \theta_2, \dots, \theta_N\}$. For each bus $\theta_i \in \mathcal{B}$, let V_i denote the voltage magnitude and θ_i the phase angle. The active and reactive power injections are defined as AC_i and Re_i respectively. Under AC power flow conditions, the power injections are modelled as:

$$AC_i = \sum_{j=1}^N V_i V_j (G_{ij} \cos(\theta_i - \theta_j) + \mathcal{B}_{ij} \sin(\theta_i - \theta_j)) \quad (47)$$

$$Re_i = \sum_{j=1}^N V_i V_j (G_{ij} \sin(\theta_i - \theta_j) - \mathcal{B}_{ij} \cos(\theta_i - \theta_j)) \quad (48)$$

Where G_{ij} and $\mathcal{B}_{ij}$ denote the conductance and susceptance between buses i and j , respectively. To simplify computation under small-angle approximations and near-unity voltages, the DC approximation is applied, yielding:

$$AC_i \approx \sum_{j \neq i} \mathcal{B}_{ij} (\theta_i - \theta_j) \quad (49)$$

$$AC = -\mathcal{B}_{\theta} \vartheta \quad (50)$$

$\mathcal{B}_{\theta}$ is the susceptance matrix, and ϑ is the vector of voltage angles. The inverse relation is given by:

$$\vartheta = -\mathcal{B}_{\theta}^{-1} AC = \mathcal{T} AC \quad (51)$$

Where $\mathcal{T}$ denotes the voltage sensitivity matrix. Let PG_i be the power produced by a generator at the bus i , Mg_i be the microgrid generation, and $demand_i$ the demand for that bus. The total power injection is:

$$AC_i = PG_i + Mg_i - demand_i \quad (52)$$

Microgrid (followers) adapt to generator output and minimize their own cost, defined as:

$$\mathcal{C}_j(Mg_j, \vartheta_j) = c_j Mg_j - Price(demand_j - Mg_j) + \frac{1}{2} Co_{\beta_j} \vartheta_j^2 \quad (53)$$

Where c_j is the cost of microgrid generation, ($Price$) the unit energy price, and Co_{β_j} the angle stability coefficient. By substituting the DC approximation into the microgrid cost, we obtain:

$$\vartheta_j = \sum_{k=1}^N \mathcal{T}_{jk} (PG_i + Mg_k - demand_k) \quad (54)$$

$$\mathcal{C}_j = c_j Mg_j - Price(demand_j - Mg_j) + \frac{1}{2} Co_{\beta_j} \left(\sum_{k=1}^N \mathcal{T}_{jk} (PG_i + Mg_k - demand_k) \right)^2 \quad (55)$$

The best response function for each microgrid is derived from first-order conditions:

$$\frac{d\mathcal{C}_j}{dMg_j} = c_j + Co_{\beta_j} \sum_{k=1}^N \mathcal{T}_{jk} \cdot \mathcal{T}_{jk} (PG_i + Mg_k - demand_k) - Price = 0. \quad (56)$$

Letting $\varphi_j = \sum_{k \neq j} \mathcal{T}_{jk} (PG_i + Mg_k - demand_k)$, and solving for Mg_j , we get:

$$Mg_j^* = \frac{1}{\mathcal{T}_{jj}} \left(\frac{Price - c_j}{Co_{\beta_j} \mathcal{T}_{jj}} - \varphi_j \right) \quad (57)$$

The strategy is bounded:

$$Mg_j^* = \max\left(0, \min\left(Mg_j^{max}, \frac{1}{\tau_{jj}}\left(\frac{Price - c_j}{Co_{\beta_j} \tau_{jj}} - \varphi_j\right)\right)\right) \quad (58)$$

This defines the Nash equilibrium among followers. The leader (generator) anticipates these responses and solves:

$$\min_{Mg, \theta} \sum_{i \in PG} (a_i PG_i^2 + b_i PG_i + \frac{1}{2} \theta_i^2) \quad (59)$$

$$PG = \mathcal{T}_1 \theta_{PG} + \mathcal{T}_2 v \quad (60)$$

Where v represents the microgrid best-response vector, and $\mathcal{T}_1, \mathcal{T}_2$ arise from the decoupled Laplacian system.

With the KKT conditions:

$$a_i PG_i + b_i - \psi_i = 0 \quad (61)$$

$$Co_{\alpha_i} \theta_i + (\mathcal{T}_1^T \psi)_i = 0 \quad (62)$$

To allow real-time grid responsiveness, we embed the Guide-Waterwheel Plant Algorithm, which adapts sensitivity coefficients using voltage rate-of-change:

$$Co_{\beta_j}^{t+1} = Co_{\beta_j}^t + \eta \left(\frac{d\theta_j}{dt}\right)^2 \quad (63)$$

$$\tau_{jk}^{(t+1)} = \tau_{jk}^{(t)} \cdot e^{-k|AC_i^{(t)} - AC_i^{(t-1)}|} \quad (64)$$

$$a_i^{(t+1)} = a_i^{(t)} + (\theta_i^{(t)} - \bar{\theta})^2 \quad (65)$$

This Stackelberg-GWPA framework ensures stability, decentralization, and adaptability under complex smart grid scenarios. Its layered structure, analytical equilibrium guarantees, and dynamic feedback yield a robust decision-making architecture suitable for real-world deployment.

Algorithm 2: Optimization of Network Dynamics

Input: Smart grid parameters (voltage, current, power flow)
Renewable energy source data (wind, solar)
Load demand data

Output: Optimized grid settings (power flow, voltage regulation)

1. **Initialize:**
2. Set up the Stackelberg game-theoretic framework
3. Initialize parameters for Guide-WWPA
4. Iterate until convergence or maximum iterations:
5. The leader proposes power dispatch
6. Followers respond with their optimal production levels
7. Calculate overall grid cost/efficiency based on the responses
8. Use Guide-WWPA to adjust power flow and voltage regulation parameters

9. Update the Leader's proposal based on Guide-WWPA and grid cost
 10. **End** Iterate
 11. **Return:** Optimized grid settings
-

Algorithm 2 illustrates the pseudocode for the proposed Stackelberg game-theoretic framework that integrates the guide-waterwheel plant algorithm (Guide-WWPA). The model incorporates the interactions between the grid operator and possible energy sources, in addition to the desired objectives of maximized power flow; voltage regulation; and more timely and successful reactions to both grid dynamics and energy losses associated with transmission. The Guide-WWPA describes an optimized procedure of modifying the grid parameters and other variables to maximize the regulation of energy flows in a Stackelberg game.

6. Control Strategy proposed

We are proposing a method that employs a convergence strategy that combines Primal-Dual and Distributed Averaging Proportional-Integral (PD-DAP) protocols to improve current regulation. We are using protocols that create a method of current optimization that will have less delay, and therefore exhibit better stability. The Primal-Dual includes cost functions and operating constraints to provide better control of current with a feedback signal when optimizing the current in real-time. This improved control allows for faster response times, reduced possibility of instability, and improved reliability in a power system. The goal is to optimize the current injections $\mathfrak{I}_i(t)$ at each node i in the network while satisfying operational constraints and minimizing a cost function related to current usage or losses. The current regulation problem is formulated as a constrained optimization problem;

$$\min_{\mathfrak{I}(t)} J(\mathfrak{I}(t)) = \sum_{i=1}^N \mathfrak{C}_i(\mathfrak{I}) \quad (66)$$

Subject to

$$\mathcal{A}\mathfrak{I}(t) = Load_{demand}, \mathfrak{I}_i^{min} \leq \mathfrak{I}_i \leq \mathfrak{I}_i^{max}, \forall i, \quad (67)$$

Where $\mathfrak{I}(t) = [\mathfrak{I}_1(t), \dots, \mathfrak{I}_N(t)]$ is the vector of current injections at the time t , $\mathfrak{C}_i(\cdot)$ are the convex cost functions related to operational cost or losses, $\mathcal{A}$ represent the network's physical laws, and $Load_{demand}$ represent load demands or other equality constraints. To solve this, we need to form the Lagrangian, so we introduced dual variables $\psi(t)$ as the agrarians on constraints:

$$\mathfrak{L}(\mathfrak{I}, \psi) = \sum_{i=1}^N \mathfrak{C}_i(\mathfrak{I}_i) + \psi(\mathcal{A}\mathfrak{I} - Load_{demand}). \quad (68)$$

The optimal solution satisfies the Karush-Kuhn-Tucker (KKT) conditions:

$$\nabla_{\mathfrak{I}_i} \mathfrak{C}_i(\mathfrak{I}_i^*) + \mathfrak{a}_i \psi^* = 0, \quad (69)$$

$$\mathcal{A}\mathfrak{X}^* = Load_{demand}, \quad (70)$$

Where $\mathfrak{a}_i$ is the i^{th} row of matrix $\mathcal{A}$.

1. Primal-Dual Algorithm

A gradient-based primal-dual iterative method updates primal and dual variables as:

$$\mathfrak{X}_i(t+1) = \mathfrak{X}_i(t) - \varrho(\nabla_{\mathfrak{X}_i} \mathfrak{C}_i(\mathfrak{X}_i(t)) + \mathfrak{a}_i \psi(t)), \quad (71)$$

$$\psi(t+1) = \psi(t) + \zeta(\mathcal{A}\mathfrak{X}(t) - Load_{demand}), \quad (72)$$

Where $\varrho, \zeta > 0$ are step sizes controlling the relative update speed. This method guarantees that the primal variables minimize cost whereas the dual variables control the network constraints with feedback allowing for real-time current control.

2. Distributed averaging proportional-integral (DAPI) control

Direct centralized implementation is impractical for large networks; thus, we use distributed averaging to share information locally. Consider a connected communication graph $\mathcal{G}$ with Laplacian matrix $\mathbb{L}$. The dual variable vector ψ is updated via distributed consensus with proportional and integral terms:

$$\begin{aligned} \psi(t) = & -k_{pr} \mathbb{L} \psi(t) - k_{In} \\ & \int_0^t \mathbb{L} \psi(\tau) d\tau + k_{co} (\mathcal{A}\mathfrak{X}(t) - Load_{demand}) \end{aligned} \quad (73)$$

Where $k_{pr}, k_{In}, k_{co} > 0$ are proportional, integral, and constraint gains. The PA-DAPI controller guarantees that the dual variables $\psi(t)$ reach consensus among the nodes, which increases stability and reduces oscillations from latency in communication or changes to the network.

V. Results and discussion

The proposed fault monitoring and prevention framework was implemented and tested using MATLAB R2023a on a computer with an Intel i5 (or equivalent) processor and at least 8 GB of RAM. The smart grid system was modelled using Simulink, and fault situations were simulated to evaluate the system.

The GAN-GDA-FLFD methodology was implemented for fault prediction and analysis utilizing the Machine Learning and Fuzzy Logic toolboxes in MATLAB. The Optimization Toolbox was used to optimize the dynamic operation of the smart grid under various conditions.

The experimental analysis of the proposed advanced monitoring, predictive analytics, and control methods is in

detail described in this section. This analysis result shows that the proposed method works well at predicting and preventing faults in a multi-source smart electrical grid. The research report, simulation settings, and comparative analysis are all included in this section. This section presents the results of the fault prediction and prevention methodology implemented in a smart electrical grid. The analysis is divided into two parts: Version 1, and Version 2. Approach, to provide a clear comparison of their performance.

- **Version 1:** A base system that integrated hybrid artificial intelligence models—IF-SVM-BWO for real-time fault forecasting and GAN-GDA-FLFD for enlightened fault avoidance.
- **Version 2:** An augmented model that further develops Version 1 through real-time network reconfiguration and backup power integration, along with active load shedding for increased fault tolerance and operational efficiency.

Both versions were tested under six energy deployment scenarios employing combinations of photovoltaic (PV), wind, and diesel sources. These scenarios were designed to account for the variability in source behavior and fluctuations in demand similar to those encountered in the real world. The simulations were primarily focused on the qualitative behavior of the framework in terms of fault identification and fault avoidance while managing the dynamic interrelationships among the heterogeneous energy sources.

1. Version 1: Using fuzzy logic (GAN-GDA-FLFD) for fault prediction and detection

Scenario 1: Photovoltaic Source (PV)

This scenario evaluated how the framework dealt with intermittency related to solar energy usage. The AI modules allowed for continuous operation by automatically observing solar input changes as they become available and consequently managing changes in the behavior of the system to avoid unexpected results.

Scenario 2: Wind Power Source

Wind-based generation presented unpredictable fluctuations in energy input. The simulation confirmed strong responsiveness to turbulence and that the model effectively ensured grid stability via predictive forecasting and heuristic fault prevention.

Scenario 3: Diesel Source

This scenario examined baseline performance under steady supply circumstances while operating on a traditional source of energy. It verified consistency in predictive methodology for monitoring in the framework and validated the flexibility of the framework for traditional generation processes.

Scenario 4: PV and Wind sources

This hybrid renewable scenario also introduced compound variability. The system control operated under synchronized reliability to control energy flow facilitated engaged and

coordinated source interaction to avoid potential defaults from interfering dual sources.

Scenario 5: PV and Diesel Sources

Using unstable and stable sources, this scenario was tested to see how well the program shifts among inputs. The smart controllers monitored load and distributed it dynamically while being able to deliver continuous and fault-tolerant power.

Scenario 6: PV, Diesel and Wind Sources

This scenario mimicked a fully functioning smart grid setup. The system was able to manage energy flows between all resources and coordinated transitions and fault events to maintain the operation of the entire system as a whole.

2. Version 2: Enhanced Grid Intelligence with Reconfiguration and Backup Management.

Scenario 1: Photovoltaic source (PV)

This version extended beyond the solar-only installation and introduced additional real-time load redirection and backup activation functions. The model retained the ability to change energy pathways when solar irradiance decreased to enable continual operation.

Scenario 2: Wind Power Source

Through adapting network-level efficiency, instability due to variations in wind was controlled through energy re-routing and adjustments to the control settings. The enhancements to response dynamics allowed for the reliability of the integrated system to function efficiently in a variable state.

Scenario 3: Diesel Source

The system provided the use of adaptive control paired with diesel input for disturbances on the grid. The model indicated that response time and fault localization improved through utilizing optimal current-based control strategies.

Scenario 4: PV and Wind Sources

The dynamic interaction between PV and wind was strengthened through smart reconfiguration mechanisms that allowed the model to sense conflicts in generation and balance source participation according to availability.

Scenario 5: PV and Diesel Sources

This scenario assessed the effectiveness of our framework in basis switching between variable and conventional sources. The power continuity has improved, response delays were lower, and the grid behaved well during transitions.

Scenario 6: PV, Diesel and Wind Sources

In this extremely dynamic grid experience, the enhanced version achieved harmonious coordination with all sources. Intelligent switching, distributed control, and real-time analytics helped the system maintain the continuity of service and adaptability of fault resilience.

Version 1: Version 1 with IF-SVM-BWO and NFT-GDA-FLFD is presented in Table 2 for six energy scenarios. The PV-only scenario (Scenario 1) resulted in a current of 1.74A, a voltage of 230 V, a state of charge (SOC) of 99.88%, a fault prediction accuracy of 95%, and a 3% loss. The wind power scenario (Scenario 2) was slightly less stable than the PV-only scenario, but it still resulted in a prediction accuracy of 95%, and the wind scenario was very dependent on high wind. The diesel generation scenario (Scenario 3) showed a lower current with a mean of 0.69A with a lower prediction accuracy of 94%. The hybrid scenarios demonstrated effective management of the multiple sources. The PV + Wind scenario (Scenario 4) gave a state of charge prediction of 94% accuracy with mean power consumed < 2.5A, and the PV + Wind + Diesel scenario (Scenario 6) reported a fault prediction accuracy of 95% but also had slightly higher transmission loss of 4.5%. Version 1: photovoltaic source with Diesel source and Wind source simulation model is shown in Fig. 4.

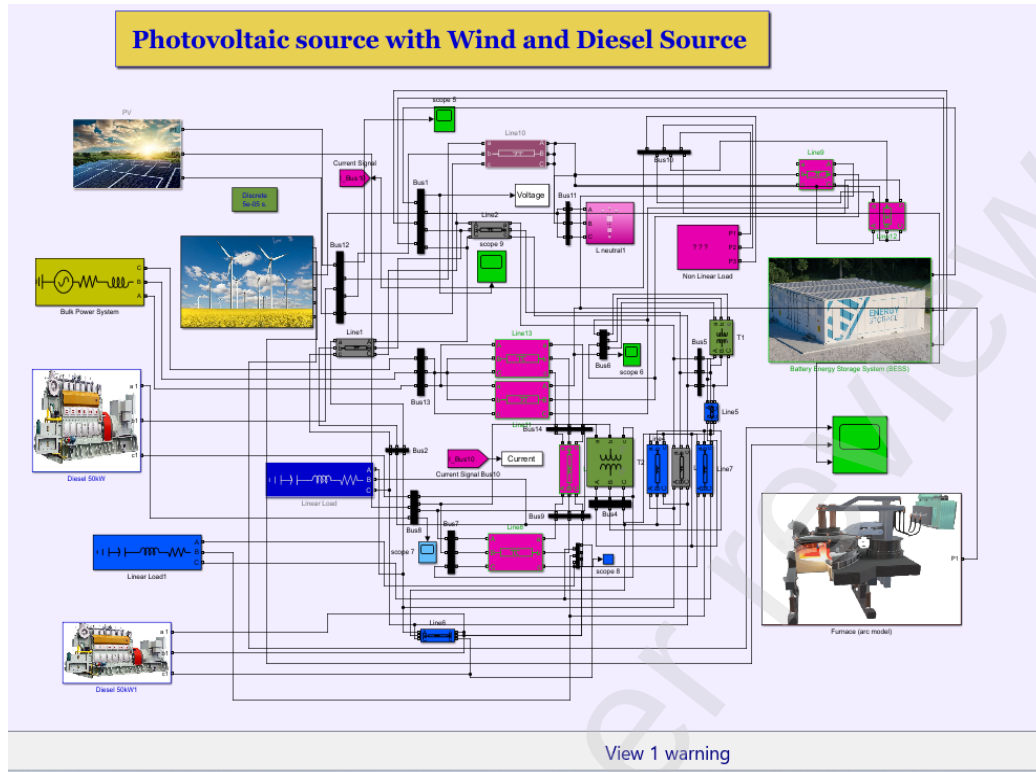


Fig. 4: Version 1: photovoltaic source with Diesel source and Wind source simulation model

Table 2: Performance Analysis of Version 1 Across Six Scenarios

Scenarios	Current (A)	Voltage (V)	SOC (%)	Fault prediction (%)	Loss (%)
Scenario 1 (Photovoltaic source)	1.74	230	99.88	95	3
Scenario 2 (Wind Power source)	1.7	230	99.88	95	3
Scenario 3 (Diesel source)	0.69	230	99.91	94	3
Scenario 4 (Photovoltaic source with Wind Power source)	1.18	230	99.9	94	3.5
Scenario 5 (Photovoltaic source with Diesel source)	1.57	230	99	95.2	4
Scenario 6 (Photovoltaic source with Diesel source and Wind source)	0.7	230	99.95	95	4.5

Version 2: Table 3 illustrates improvements to Version 2 as a result of real-time reconfiguration and backup integrated into the modified advanced hybrid system. In PV-only operation (Scenario 1), 2A at 200V was handled with a SOC of 96.99% and prediction accuracy of 95%. In the wind scenario (Scenario 2), it remained at SOC 99.7% with 85% prediction accuracy on extreme behavior. Diesel (Scenario 3) was optimized to 1.5A/140V with 95% prediction accuracy. The full hybrid scenario (Scenario 6) demonstrates the maximum performance achieved under the new multi-source adaptive scheme at 99%

of all fault predictions and a relatively consistent 3% loss across all scenarios, showing that improved grid behavior and fault tolerance have been established. Version 2: photovoltaic source with Diesel source and Wind source simulation model is shown in Fig. 5. Version 2 thus exhibited quantifiable advancements over Version 1. Particularly in hybrid scenarios, fault prediction increased from 95% to 99% and transmission loss decreased from 4.5% to 3%. There were no differences between version 1 and version 2 for voltage outputs, but current regulation and SOC management were better in version 2. In Scenario 6, SOC improved (noted as a negative change) from 99.95% in version

1 and was improved to 99.8% in version 2. The increased resource optimization capability of the enhanced system was particularly effective in maintaining a 3% transmission loss

across all the scenarios while version 1 exhibited a range of 3-4.5%.

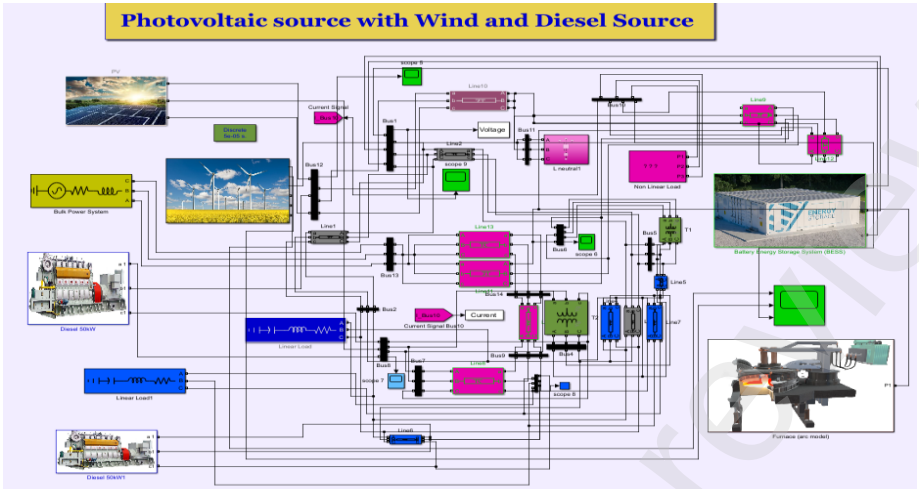


Fig. 5: Version 2: photovoltaic source with Diesel source and Wind source simulation model

Table 3: Performance Analysis of Version 2 Across Six Scenarios

Scenarios	Current (A)	Voltage (V)	SOC (%)	Fault prediction (%)	Loss (%)
Scenario 1 (Photovoltaic source)	2	200	96.99	95	3
Scenario 2 (Wind Power source)	2	200	99.7	85	3
Scenario 3 (Diesel source)	1.5	140	99.92	95	3
Scenario 4 (Photovoltaic source with Wind Power source)	1.1	170	99.7	93	3
Scenario 5 (Photovoltaic source with Diesel source)	2	200	99.5	95	3
Scenario 6 (Photovoltaic source with Diesel source and Wind source)	2	0.7	190	99	3

3. Comparative analysis

This section provides a comparative analysis of the proposed method with existing methods in the context of Scenario 6 when the smart electrical grid uses photovoltaic energy, diesel, and wind power. Scenario 6 represents a complicated and highly applicable configuration because it includes the multitude of problems and opportunities in modern multi-source energy systems: Solar, wind, and diesel generation introduce complexities in grid management, fault detection, and overall grid optimization with a combination of intermittent generation resources and the pressure for reliable back-up generation when renewables are intermittent. This section compares the performance of the SDTF-Guide-WWPA and PD-DAP1 protocols (the proposed method) to existing methods in Scenario 6 (Photovoltaic source added Diesel and Wind

sources), to assess the implementation of the proposed framework in a complex multi-source setting.

Existing methods included the traditional optimization technique of MLP-ELM (Multi-Layer Perceptron with Extreme Learning Machine). The evaluation of performance was compared according to the following metrics: Current (A), Voltage (V), State of Charge (SOC), Fault Prediction Accuracy (%), and Transmission Loss (%).

1. Performance of the Current (A)

Fig. 6 depicts the current (in amperes) with time, which is a critical metric in battery systems. To identify system flaws, precise current sensing is necessary. Current fluctuations could be a sign of sensor problems or battery problems. The following expression is given:

$$\mathfrak{T}(t) = \frac{\mathbb{V}(t)}{\mathfrak{R}} \quad (74)$$

The current time t is denoted by $\mathfrak{T}(t)$, the voltage by $\mathbb{V}(t)$, and the resistance (assumed constant) by $\mathfrak{R}$.

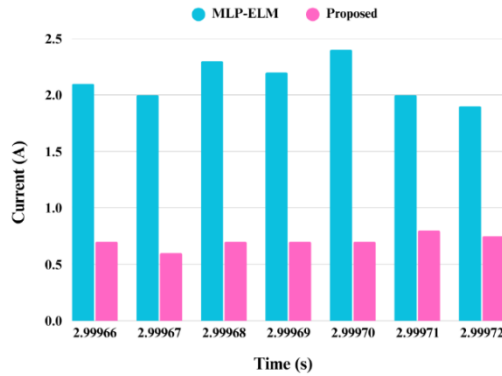


Fig. 6 the evolution of the current

TABLE 4: Numerical Outcomes of Time vs. Current

Time (s) x-axis	MLP-ELM	Proposed
2.99966	2.1	0.7
2.99967	2.0	0.6
2.99968	2.3	0.7
2.99969	2.2	0.7
2.99970	2.4	0.7
2.99971	2.0	0.8
2.99972	1.9	0.75

Current is an essential parameter for smart grids, as it directly represents the operational load along with the health of the power system. In Table 4, we can see the conventional MLP-ELM method displays higher current values, ranging statistically from 1.9 A to 2.4 A between the different unique instances. This leads to the conclusion that not only does the MLP-ELM method have a higher average current, but it is also an unstable load that cannot regulate energy efficiently for balancing control and interaction. The proposed PD-DAPI and SDTF-Guide-WWPA framework demonstrated loading in a stable average of 0.7 A. This was a significant improvement over the previous methods in the sense of load-regulating behavior. Many variances seen in the proposed methods results can be observed in Fig. 6 above.

2. Performance of the Voltage (V)

For connected devices to operate reliably, the voltage must be kept within acceptable levels for customers and the stability of the grid. The Time vs. Voltage graph illustrates voltage over time for Scenario 6 looking at the proposed and existing methods.

$$\mathbb{V}(t) = \mathbb{V}_0 + \mathfrak{T}R_{esist}(t) \quad (75)$$

where $R_{esist}(t)$ stands for time-varying resistance, and $\mathbb{V}_0$ for beginning voltage.

TABLE 5: Numerical outcomes of Time vs. Voltage (V)

Time (s) x-axis	MLP-ELM	Proposed
2.99966	210	230
2.99967	205	220
2.99968	215	230
2.99969	200	210
2.99970	195	200
2.99971	205	220
2.99972	201	230

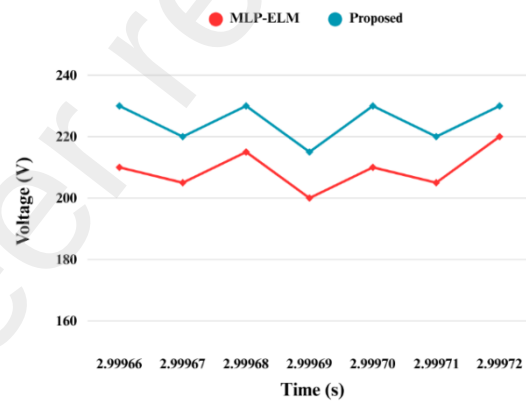


Fig. 7: The evolution of the voltage

Voltage stability ensures the safe and reliable operation of electrical equipment that is connected to the grid. In Table 5, the MLP-ELM method has voltage values between 195V to 215V and showed bad regulation under dynamic conditions. The proposed model has voltage profiles at 230V, showing resistance to changes in voltage profiles, as noted in Fig. 7 (Time vs. Voltage). The sustained voltage profile demonstrates the efficiency of the Guide-WWPA algorithm in voltage regulation and helps reduce the impacts of delays with the DAPI control mechanism. The proposed method's ability to maintain voltage at the nearest optimal level helps improve the overall power quality, reducing the chance of under-voltage (low voltage) and over-voltage (high voltage) damage to sensitive grid components.

3. Performance of the SOC (%)

A percentage representing the battery's remaining capacity is displayed by the state of charge. The lifespan of the battery and the avoidance of malfunctions depend on regular monitoring of the SoC.

$$SOC(t) = SOC_0 - \int_0^t \frac{\mathfrak{T}(t) \cdot \Delta t}{C_{batt}} \quad (76)$$

Where C_{batt} : capacity of the battery

TABLE 6: Numerical outcomes of Time vs. SOC (%)

Time (s) x-axis	MLP-ELM	Proposed
2.99966	98.5	99.1
2.99967	98.4	99.2
2.99968	98.3	99.3
2.99969	98.2	99.4
2.99970	98.1	99.5
2.99971	98	99.6
2.99972	97.5	99.8

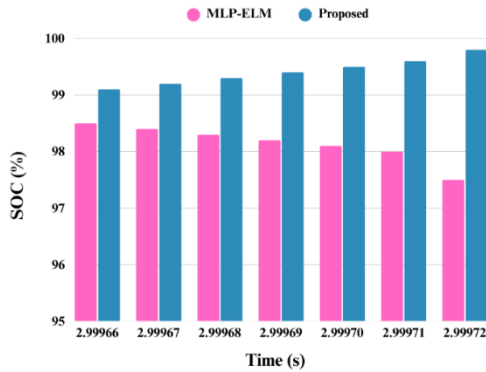


Fig. 8: The evolution of the SOC (%)

SOC (State of Charge) is representative of remaining energy in storage systems and consequential for energy management, particularly in hybrid renewables models. The dynamic/reduction keeps with results from the MLP-ELM model, indicating that the SOC moves gradually downward; from 98.5% to 97.5%, it shows energy inefficiency as seen in Table 4. In a similar vein, the SOC of the proposed model is in the opposite fashion from 99.1% to 99.8%. It was shown that the time spent in the storage section is improved due to the stored energy being used rather than decreasing for later use/more optimal energy use, and home and storage were better. Fig. 8 and Table 6 effectively illustrate the increase in SOC as the proposed system progresses. The interpretation of results indicates the realized ability to effectively, optimally, and automatically utilize backup sources, real-time reconfiguration, and re-optimizing loads to bring improved autonomy, robustness, and necessary levels during high demand.

4. The evolution of the fault prediction (%)

The fault diagnosis over time graph shows how fast and correctly the system can identify and respond to sensor or

battery failures. More efficient system maintenance results from quicker detection.

$$Accuracy(t) = \frac{TP(t) + TN(t)}{TP(t) + TN(t) + FP(t) + FN(t)} \times 100 \quad (77)$$

True positives $TP(t)$, true negatives $TN(t)$, false positives $FP(t)$, and false negatives $FN(t)$ are used to calculate accuracy. The IF-SVM-BWO model uses improved hyperparameters to lower $FN(t)$ and $FP(t)$ rates.

TABLE 7: Numerical outcomes of Time vs. Fault prediction (%)

Time (s) x-axis	MLP-ELM	Proposed
2.99966	83	95
2.99967	84	94.5
2.99968	85	95
2.99969	86	95.5
2.99970	87	95
2.99971	88	94
2.99972	89	96

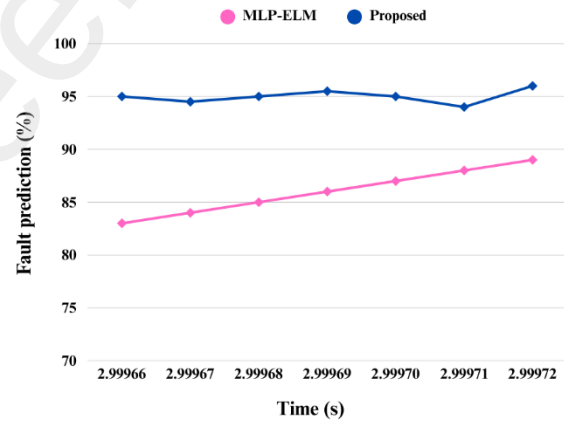


Fig. 9: The evolution of the fault prediction (%)

Accurate fault prediction is crucial for the avoidance of unexpected outages and increased safety while on the job. From Table 7, we can see a range of accuracy for the MLP-ELM system from 83% to 89%, while the proposed IF-SVM-BWO model did achieve a high of 96% accuracy. Fig. 9 displays the proposed system for time vs fault prediction accuracy as performed far better in both time and accuracy, presenting a smoother and better prediction curve. We can also highlight that using additional optimization techniques and hybrid AI models in the traditional trajectory in addition to the GAN and fuzzy logic features offered the system a unique opportunity to develop a dynamic capacity for the simulation of potential faults, detection, and mitigation of faults even under varied grid scenarios, and we would classify the system as robust.

5. Number of Epochs vs Loss (%)

This analysis compares the training performance of the proposed and existing methods by graphing the relationship between the number of training epochs and the relevant loss percentage.

$$Loss(e) = \alpha, \beta, \gamma \quad (78)$$

where α, β, γ are optimization constants.

TABLE 8: Numerical outcomes of Epochs vs. Loss (%)

Epochs/Iteration x-axis	MLP-ELM	Proposed
10	8.5	5.0
20	7.2	4.5
30	6.0	4.0
40	5.5	3.8
50	5.2	3.5
60	5.0	3.2
70	4.8	3.0

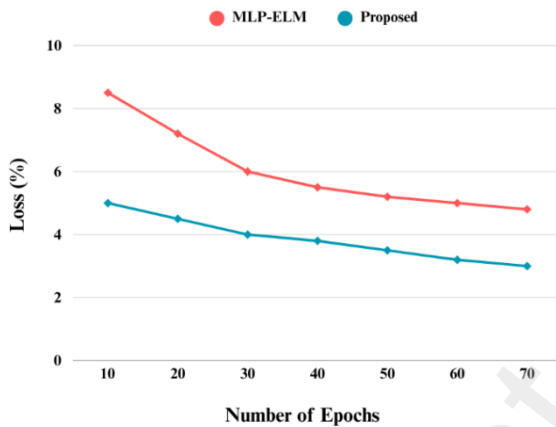


Fig. 10 Epochs vs Loss (%)

Transmission loss constitutes loss of energy in the system and consequently, a contribution to operational inefficiency. From the information in Table 8, we observe that although MLP-ELM decreases loss from 8.5% to 4.8%, the proposed method is more efficient with a loss of only 3.0%. This behavior was further corroborated by Fig. 10 Epochs vs Loss, where the proposed model reached faster convergence and had a lower loss value at the end. The information demonstrates that the PD-DAPI and Guide-WWPA combined method is capable of not only real-time dynamic optimization but also reducing losses related to energy loss during prolonged grid operations. Reducing loss signifies that a larger portion of power generation is utilized, thus less environmental impact, and decreased operating costs.

Initially, we constructed a smart electrical grid with 14 Buses, PV with Diesel and wind sources, and BESS in the Simulink Model based on scenarios. Then, we collect voltage, current, power, and temperature data and preprocess using

noise reduction and normalization techniques. Next, we monitor the current leakage using Particle Swarm Optimization and Genetic Algorithm (PSO-GA). Next, we predict the faults using IF-SVM with the Blue Whale (BWO) Optimization technique for efficient prediction. Next, we prevent the faults using Gaussian Discriminant Analysis and Fuzzy Logic Based Fault Detection (GAN-GDA-FLFD) method. Next, we optimize the data using the Stackelberg game-theoretic framework with the Guide-Waterwheel Plant Algorithm (SDTF-Guide-WWPA).

Next, we implement Primal-Dual and Distributed Averaging Proportional-Integral (PD-DAPI) protocols to control current-based regulation. Finally, we processed performance metrics plots for the following: Time vs Current (A), Time vs Voltage (V), Time vs SOC (%), Time vs Fault prediction (%), and Number of Epochs/Iterations vs Loss (%). In fig 6-10 and table 4-8 illustrates the performance metrics.

VI. CONCLUSION

The rise in complexity and interoperability of renewable energy sources into smart grids has led to the demand for next-generation solutions for fault prediction, prevention, and dynamic optimization. This research explored key issues in leakage current monitoring, fault detection reliability, adaptive optimization of grid operation, and stability of control using a range of novel methodologies. The proposed framework made possible the integration of hybrid algorithms for this purpose including PSO-GA for insulation degradation monitoring, IF-SVM-BWO for online fault classification, and GAN-GDA-FLFD for pre-emptive fault mitigation, which supported improvements in grid resilience and grid operations. The Stackelberg game-theoretic framework, augmented with the Guide-WWPA algorithm as an intervention plan, optimized power flow and voltage compliance, while the PD-DAPI protocol optimized the slit by mitigating latencies and optimizing stability concerning current-based control, to then drive changes. The test of the full framework in six multi-source scenarios provided experimental evidence for the effectiveness of the framework, with fault prediction accuracies and success rates of up to 99% and reduced transmission losses of up to 3% in hybrid configurations. The implications highlight the persistent opportunity of sophisticated, data-driven systems to provide systems greater resiliency in their ability to manage renewable penetrations and proactively engage with operational uncertainties in the process of managing grids at scale.

References

1. Rolo, I., Costa, V. A., & Brito, F. P. (2023). Hydrogen-based energy systems: Current technology development status, opportunities and challenges. *Energies*, 17(1), 180.
2. Bakare, M. S., Abdulkarim, A., Zeeshan, M., & Shuaibu, A. N. (2023). A comprehensive overview on demand side energy management towards smart grids:

challenges, solutions, and future direction. *Energy Informatics*, 6(1), 4.

3. Perez, M., & Perez, R. (2022). Update 2022—A fundamental look at supply side energy reserves for the planet. *Solar Energy Advances*, 2, 100014.
4. Fang, X. Energy Equity-Aware Load Shedding Optimisation Methodology. *Energy Internet*, e12027.
5. Chanchangi, Y. N., Adu, F., Ghosh, A., Sundaram, S., & Mallick, T. K. (2023). Nigeria's energy review: Focusing on solar energy potential and penetration. *Environment, Development and Sustainability*, 25(7), 5755-5796.
6. Saleem, M. U., Usman, M. R., Usman, M. A., & Politis, C. (2022). Design, deployment and performance evaluation of an IoT based smart energy management system for demand side management in smart grid. *IEEE Access*, 10, 15261-15278.
7. Carr, D., & Thomson, M. (2022). Non-technical electricity losses. *Energies*, 15(6), 2218.
8. Štogl, O., Miltner, M., Zanolco, C., Traverso, M., & Starý, O. (2024). Electric vehicles as facilitators of grid stability and flexibility: A multidisciplinary overview. *Wiley Interdisciplinary Reviews: Energy and Environment*, 13(5), e536.
9. Ahmed, M. M. R., Mirsaeidi, S., Koondhar, M. A., Karami, N., Tag-Eldin, E. M., Ghamry, N. A., & Sharaf, A. M. (2024). Mitigating uncertainty problems of renewable energy resources through efficient integration of hybrid solar PV/wind systems into power networks. *IEEE Access*, 12, 30311-30328.
10. Blaabjerg, F., Yang, Y., Kim, K. A., & Rodriguez, J. (2023). Power electronics technology for large-scale renewable energy generation. *Proceedings of the IEEE*, 111(4), 335-355.
11. Verma, N., Kumar, N., Gupta, S., Malik, H., & Márquez, F. P. G. (2023). Review of sub-synchronous interaction in wind integrated power systems: classification, challenges, and mitigation techniques. *Protection and Control of Modern Power Systems*, 8(2), 1-26.
12. Lunardi, A., Normandia Lourenço, L. F., Munkhchuluun, E., Meegahapola, L., & Sguarezi Filho, A. J. (2022). Grid-connected power converters: An overview of control strategies for renewable energy. *Energies*, 15(11), 4151.
13. Panda, S., Mohanty, S., Rout, P. K., Sahu, B. K., Parida, S. M., Kotb, H., & Shouran, M. (2022). An insight into the integration of distributed energy resources and energy storage systems with smart distribution networks using demand-side management. *Applied Sciences*, 12(17), 8914.
14. Cavus, M. (2025). Advancing Power Systems with Renewable Energy and Intelligent Technologies: A Comprehensive Review on Grid Transformation and Integration. *Electronics*, 14(6), 1159.
15. Rezaeimozafer, M., Monaghan, R. F., Barrett, E., & Duffy, M. (2022). A review of behind-the-meter energy storage systems in smart grids. *Renewable and Sustainable Energy Reviews*, 164, 112573.
16. Wu, B., & Hu, Y. (2023). Analysis of substation joint safety control system and model based on multi-source heterogeneous data fusion. *IEEE Access*, 11, 35281-35297.
17. Xiao, J., Wang, Y., Zhang, X., Luo, G., & Xu, C. (2024). Multi source data security protection of smart grid based on edge computing. *Measurement: Sensors*, 35, 101288.
18. Zhou, H., Liang, J., Du, X., & Wu, M. (2024). Multi-Timescale Reactive Power Optimization and Regulation Method for Distribution Networks Under a Multi-Source Interaction Environment. *Processes*, 12(10), 2254.
19. Li, Q., Deng, Y., Liu, X., Sun, W., Li, W., Li, J., & Liu, Z. (2023). Autonomous smart grid fault detection. *IEEE Communications Standards Magazine*, 7(2), 40-47.
20. Li, B., Liao, Y., Liu, S., Liu, C., & Wu, Z. (2025). Research on Short-Term Load Forecasting of LSTM Regional Power Grid Based on Multi-Source Parameter Coupling. *Energies*, 18(3), 516.
21. Wu, Y., Wang, Q., Guo, N., Tian, Y., Li, F., & Su, X. (2023). Efficient multi-source self-attention data fusion for fdia detection in smart grid. *Symmetry*, 15(5), 1019.
22. Tavakoli, S., Zamani, A. A., & Khajehoddin, A. (2024). Efficient load frequency control in multi-source interconnected power systems using an innovative intelligent control framework. *Energy Reports*, 11, 2805-2817.
23. Wang, L., He, Y., Liu, X., Li, L., & Shao, K. (2022). M2TNet: Multi-modal multi-task Transformer

network for ultra-short-term wind power multi-step forecasting. *Energy Reports*, 8, 7628-7642.

24. Lepolesa, L. J., Achari, S., & Cheng, L. (2022). Electricity theft detection in smart grids based on deep neural network. *Ieee Access*, 10, 39638-39655.
25. Pan, X., Chen, M., Zhao, T., & Low, S. H. (2022). DeepOPF: A feasibility-optimized deep neural network approach for AC optimal power flow problems. *IEEE Systems Journal*, 17(1), 673-683.
26. Mystakidis, A., Ntozi, E., Afentoulis, K., Koukaras, P., Gkaidatzis, P., Ioannidis, D., & Tzovaras, D. (2023). Energy generation forecasting: elevating performance with machine and deep learning. *Computing*, 105(8), 1623-1645.
27. Tiwari, D., Zideh, M. J., Talreja, V., Verma, V., Solanki, S. K., & Solanki, J. (2024). Power flow analysis using deep neural networks in three-phase unbalanced smart distribution grids. *IEEE Access*, 12, 29959-29970.
28. Ibrahim, B., Rabelo, L., Gutierrez-Franco, E., & Clavijo-Buritica, N. (2022). Machine learning for short-term load forecasting in smart grids. *Energies*, 15(21), 8079.
29. Li, Y., Cao, J., Xu, Y., Zhu, L., & Dong, Z. Y. (2024). Deep learning based on Transformer architecture for power system short-term voltage stability assessment with class imbalance. *Renewable and Sustainable Energy Reviews*, 189, 113913.
30. Li, Y., He, S., Li, Y., Ge, L., Lou, S., & Zeng, Z. (2022). Probabilistic charging power forecast of EVCS: Reinforcement learning assisted deep learning approach. *IEEE Transactions on Intelligent Vehicles*, 8(1), 344-357.
31. Jithish, J., Alangot, B., Mahalingam, N., & Yeo, K. S. (2023). Distributed anomaly detection in smart grids: a federated learning-based approach. *IEEE Access*, 11, 7157-7179.
32. Kumar, K., Pande, S. V., Kumar, T. C. A., Saini, P., Chaturvedi, A., Reddy, P. C. S., & Shah, K. B. (2023). Intelligent controller design and fault prediction using machine learning model. *International Transactions on Electrical Energy Systems*, 2023(1), 1056387.
33. Alhanaf, A. S., Balik, H. H., & Farsadi, M. (2023). Intelligent fault detection and classification schemes for smart grids based on deep neural networks. *Energies*, 16(22), 7680.
34. Bakkar, M., Bogarra, S., Córcoles, F., Iglesias, J., & Al Hanaineh, W. (2022). Multi-layer smart fault protection for secure smart grids. *IEEE Transactions on Smart Grid*, 14(4), 3125-3135.
35. Mbey, C. F., Foba Kakeu, V. J., Boum, A. T., & Souhe, F. G. Y. (2023). Fault detection and classification using deep learning method and neuro-fuzzy algorithm in a smart distribution grid. *The Journal of Engineering*, 2023(11), e12324.
36. Sopelsa Neto, N. F., Stefenon, S. F., Meyer, L. H., Ovejero, R. G., & Leithardt, V. R. Q. (2022). Fault prediction based on leakage current in contaminated insulators using enhanced time series forecasting models. *Sensors*, 22(16), 6121.
37. Tan, Y., Zhou, L., Xue, X., & Duan, B. (2023). Exploration of Key Technologies for Equipment Operation and Maintenance Based on New Power Systems. *International Journal of Thermofluids*, 20, 100482.
38. Ortiz-Matos, L., Zea, L. B. G., & González-Sanchez, J. W. (2023). A methodology of sensor fault-tolerant control on a hierarchical control for hybrid microgrids. *IEEE Access*, 11, 58078-58098.
39. Alsirhani, A., Alshahrani, M. M., Abukwaik, A., Taloba, A. I., Abd El-Aziz, R. M., & Salem, M. (2023). A novel approach to predicting the stability of the smart grid utilizing MLP-ELM technique. *Alexandria Engineering Journal*, 74, 495-508.
40. Peng, B., Ma, X., Ma, X., Tian, C., & Sun, Y. (2023). Coordinated AGC control strategy for an interconnected multi-source power system based on distributed model predictive control algorithm. *Frontiers in Energy Research*, 10, 1019464.